Homework1

Lecture1

2[Page 27]. Two six-sided dice are thrown sequentially, and the face values that come up are recorded.

- a. List the sample space.
- b. List the elements that make up the following events:
  - A = the sum of the two values is at least 5
  - B = the value of the first die is higher than the value of the second
  - C = the first value is 4
- c. List the elements of the following event
  - $A \cap C$
  - $B \cup C$
  - $A \cap (B \cup C)$

Solution: (a) Sample space:

$\{(1,1),(1,2),(1,3),(1,4),(1,5),(1,6),$
$(2,1),(2,2),(2,3),(2,4),(2,5),(2,6),$
$(3,1),(3,2),(3,3),(3,4),(3,5),(3,6),$
$(4,1),(4,2),(4,3),(4,4),(4,5),(4,6),$
$(5,1),(5,2),(5,3),(5,4),(5,5),(5,6),$
$(6,1),(6,2),(6,3),(6,4),(6,5),(6,6)\}$

(b). $A = \{(1,4),(1,5),(1,6),$
$(2,3),(2,4),(2,5),(2,6),$
$(3,2),(3,3),(3,4),(3,5),(3,6),$
$(4,1),(4,2),(4,3),(4,4),(4,5),(4,6),$
$(5,1),(5,2),(5,3),(5,4),(5,5),(5,6),$
$(6,1),(6,2),(6,3),(6,4),(6,5),(6,6)\}$

$B = \{(2,1)$
$(3,1),(3,2),$
$(4,1),(4,2),(4,3),$
$(5,1),(5,2),(5,3),(5,4),$
$(6,1),(6,2),(6,3),(6,4),(6,5)\}$

$C = \{(4,1),(4,2),(4,3),(4,4),(4,5),(4,6)\}$

(c). $A \cap C = C = \{(4,1),(4,2),(4,3),(4,4),(4,5),(4,6)\}$

$B \cup C = \{(2,1)$
$(3,1),(3,2),$
$(4,1),(4,2),(4,3),(4,4),(4,5),(4,6)$
$(5,1),(5,2),(5,3),(5,4),$
$(6,1),(6,2),(6,3),(6,4),(6,5)\}$

$A \cap (B \cup C) = \{$
$(3,2),$
$(4,1),(4,2),(4,3),(4,4),(4,5),(4,6)$
$(5,1),(5,2),(5,3),(5,4),$
$(6,1),(6,2),(6,3),(6,4),(6,5)\}$

5[Page 27]. Let A and B be arbitrary events. Let C be the event that either A occurs or B occurs, <u>but not both</u>. Express C in terms of A and B using any of the basic operations of union, intersection, and complement.

Solution: $(A \cup B) - (A \cap B)$ or $(A^c \cap B) \cup (B^c \cap A)$

$= (A \cup B) \cap (A \cap B)^c$

Leture2

7[Page 27]. Prove Bonferroni's inequality

$$P(AB) \geq P(A) + P(B) - 1 \tag{1}$$

Proof: $P(A\cap B) + P(A\cap B)^c = 1$

$\Rightarrow P(A\cap B) = 1 - P(A\cap B)^c$

De Morgan Laws

$= 1 - P(A^c \cup B^c)$

This item ≥ 0

Addition law

$= 1 - [P(A^c) + P(B^c) - P(A^cB^c)]$

$= 1 - P(A^c) - P(B^c) + P(A^cB^c)$

$\geq \underbrace{1 - P(A^c)}_{P(A)} - 1 + \underbrace{1 - P(B^c)}_{P(B)} = P(A) + P(B) - 1$

Therefore, we have $P(A\cap B) \geq P(A) + P(B) - 1$

Q.E.D.

8[Page 27]. Prove that

$$P\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} P(A_i) \tag{2}$$

Proof: We have many ways to prove this exercise.

① Mathmatical induction

② Let $B_1 = A_1$, $B_2 = A_2 - B_1$, $B_3 = A_3 - \bigcup_{i=1}^{2} B_i$, ...

$B_n = A_n - \bigcup_{i=1}^{n-1} B_i$.

Then $\bigcup_{i=1}^{\infty} A_i = \bigcup_{i=1}^{\infty} B_i$, $B_i \subseteq A_i$ and $\{B_n\}$ are pairwise disjoint.

We have $P(\bigcup_{i=1}^{\infty} A_i) = P(\bigcup_{i=1}^{\infty} B_i) = \sum_{i=1}^{\infty} P(B_i) \leq \sum_{i=1}^{\infty} P(A_i)$

Q.E.D.

Supplement Questions

1. For three random events $A, B, C, P(A) = P(B) = P(C) = \frac{1}{4}, P(AB) = P(BC) = 0, P(AC) = \frac{1}{8}$. What is the probability of at least one of events A, B, C happens?

Solution: $P(A\cup B\cup C) = P(A) + P(B) + P(C) - P(AB) - P(AC) - P(BC) + P(ABC)$

$= \frac{1}{4} \times 3 - 0 - 0 - \frac{1}{8} + 0$

Since $0 \leq P(ABC) \leq P(BC) = 0$

$P(ABC)$ must $= 0$

$$= \frac{5}{8}.$$

2. Let A, B be two events that satisfy $P(AB) = P(A^cB^c)$, and $P(A) = p$. Find $P(B)$.

Solution: $1 - P(AB) = P(\overline{AB}) = P(A^c \cup B^c)$

$$= P(A^c) + P(B^c) - P(A^cB^c)$$

$$= 1 - P(A) + 1 - P(B) - P(AB)$$

Hence $1 = 1 - P(A) + 1 - P(B)$

$$\Rightarrow P(B) = 1 - P(A) = 1 - p.$$

Leture3

12[Page 27]. In a game of poker, five players are each dealt 5 cards from a 52-card deck. How many ways are there to deal the cards?

Solution: $\binom{52}{5} \cdot \binom{47}{5} \cdot \binom{42}{5} \cdot \binom{37}{5} \cdot \binom{32}{5}$

29[Page 29]. A poker player is dealt three spades and two hearts. He discards the two hearts and draws two more cards. What is the probability that he draws two more spades?

Solution: $$\frac{\binom{10}{2}}{\binom{47}{2}}$$

Only $52 - 5 = 47$ cards

$13 - 3 = 10$ spades now

Supplement Questionssss

1. Choose $2r$ shoes from n pairs of shoes with different sizes ($2r < n$). Compute the probabilities of the following events.
   - There are no matching pairs among the $2r$ shoes.
   - There is exact **ONE** matching pair among the $2r$ shoes.
   - There are r matching pairs among the $2r$ shoes

(a) $$p = \frac{C_n^{2r} \cdot 2^{2r}}{C_{2n}^{2r}}$$

(b) $$p = \frac{C_n^{1} \cdot C_{n-1}^{2r-2} \cdot 2^{2r-2}}{C_n^{2r}}$$

(c) $$p = \frac{C_n^{r}}{C_n^{2r}}$$